

Ex-2 Cross Platform Prompting – Evaluating Diverse Techniques in AI Powered Text Summarization

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Aim

To understand and evaluate different prompting techniques across multiple AI platforms by performing text summarization and comparing the outputs.

Introduction

Artificial Intelligence (AI) platforms such as OpenAI ChatGPT, Google Gemini, and Microsoft Copilot are widely used for text summarization tasks. Different prompting techniques can significantly influence the quality and accuracy of generated summaries. This experiment evaluates how various prompting methods affect summarization performance across multiple AI platforms.

Algorithm

1. Select a common paragraph for summarization.
2. Open AI platforms such as ChatGPT, Gemini, and Copilot.

3. Provide the same input text to all platforms.
4. Apply different prompting techniques:
 - Zero-shot prompting
 - Few-shot prompting
 - Chain-of-thought prompting
 - Role-based prompting
5. Record the outputs generated by each platform.
6. Compare the outputs based on:
 - Accuracy
 - Coherence
 - Simplicity
 - Speed
 - User Experience
7. Analyze the differences between outputs.
8. Prepare a comparison table.
9. Document observations and conclusion.
10. Convert the report into PDF format and submit.

Input Text

Topic: Blockchain Technology

Blockchain technology is a decentralized digital ledger system that records transactions across multiple computers securely and transparently. Each block contains transaction data, a timestamp, and a cryptographic hash of the previous block, forming a chain. Blockchain eliminates the need for intermediaries, improves security, reduces fraud, and enhances transparency. It is widely used in cryptocurrencies, supply chain management, healthcare, banking, and smart contracts. Despite its advantages, blockchain faces challenges such as scalability, high energy consumption, and regulatory concerns.

Prompting Techniques

1. Zero-Shot Prompting

Prompt

Summarize the given paragraph about Blockchain Technology in simple words.

ChatGPT Output

Blockchain is a secure digital system used to store transaction records across many computers. It improves transparency, reduces fraud, and is used in areas like banking and healthcare.

Gemini Output

Blockchain technology is a decentralized system that securely records transactions without intermediaries. It is widely used in cryptocurrency and other industries but has some scalability and energy challenges.

Copilot Output

Blockchain is a digital ledger technology that stores information securely and transparently. It helps improve trust and security in many industries.

2. Few-Shot Prompting

Prompt

Example 1:

AI is used in healthcare to improve diagnosis and patient care.

Summary: AI helps healthcare by improving diagnosis and treatment.

Example 2:

Cloud computing allows users to access data online from anywhere.

Summary: Cloud computing enables online access to data and services.

Now summarize the following paragraph about Blockchain Technology.

ChatGPT Output

Blockchain technology is a secure and transparent system for recording transactions digitally. It is useful in many industries and helps reduce fraud and improve trust.

Gemini Output

Blockchain is a decentralized digital ledger used to store transactions securely. It supports industries like banking and healthcare while ensuring transparency.

Copilot Output

Blockchain securely records transactions using decentralized technology. It improves security, transparency, and efficiency in different sectors.

3. Chain-of-Thought Prompting

Prompt

Read the paragraph carefully. Identify the main idea, important applications, advantages, and challenges. Then generate a detailed summary.

ChatGPT Output

Blockchain technology is a decentralized and secure digital ledger that records transactions transparently across multiple systems. It removes intermediaries and improves trust, security, and fraud prevention. Blockchain is commonly used in cryptocurrency, healthcare, banking, supply chain management, and smart contracts. However, issues such as scalability, high power consumption, and government regulations remain challenges.

Gemini Output

Blockchain is a distributed ledger technology that securely stores transaction records in linked blocks. It increases transparency and security while supporting applications like cryptocurrencies and smart contracts. The major limitations include energy usage and scalability concerns.

Copilot Output

Blockchain technology provides secure and transparent transaction recording through decentralized systems. It is useful in finance, healthcare, and logistics, though challenges such as energy consumption and regulation affect its growth.

4. Role-Based Prompting

Prompt

Assume you are a technology researcher. Summarize the paragraph about Blockchain Technology for college students.

ChatGPT Output

Blockchain is a decentralized digital ledger that securely stores transaction data in connected blocks. It improves transparency, trust, and security by removing intermediaries. Blockchain has applications in cryptocurrency, banking, healthcare, and supply chains, but challenges like scalability and energy usage must be addressed.

Gemini Output

As a researcher, blockchain can be described as a secure and transparent transaction system that works without central control. It has many real-world applications but faces technical and regulatory limitations.

Copilot Output

Blockchain is a modern technology that records transactions safely and transparently across many computers. It supports industries such as banking and healthcare while improving trust and reducing fraud.

Comparative Analysis Table

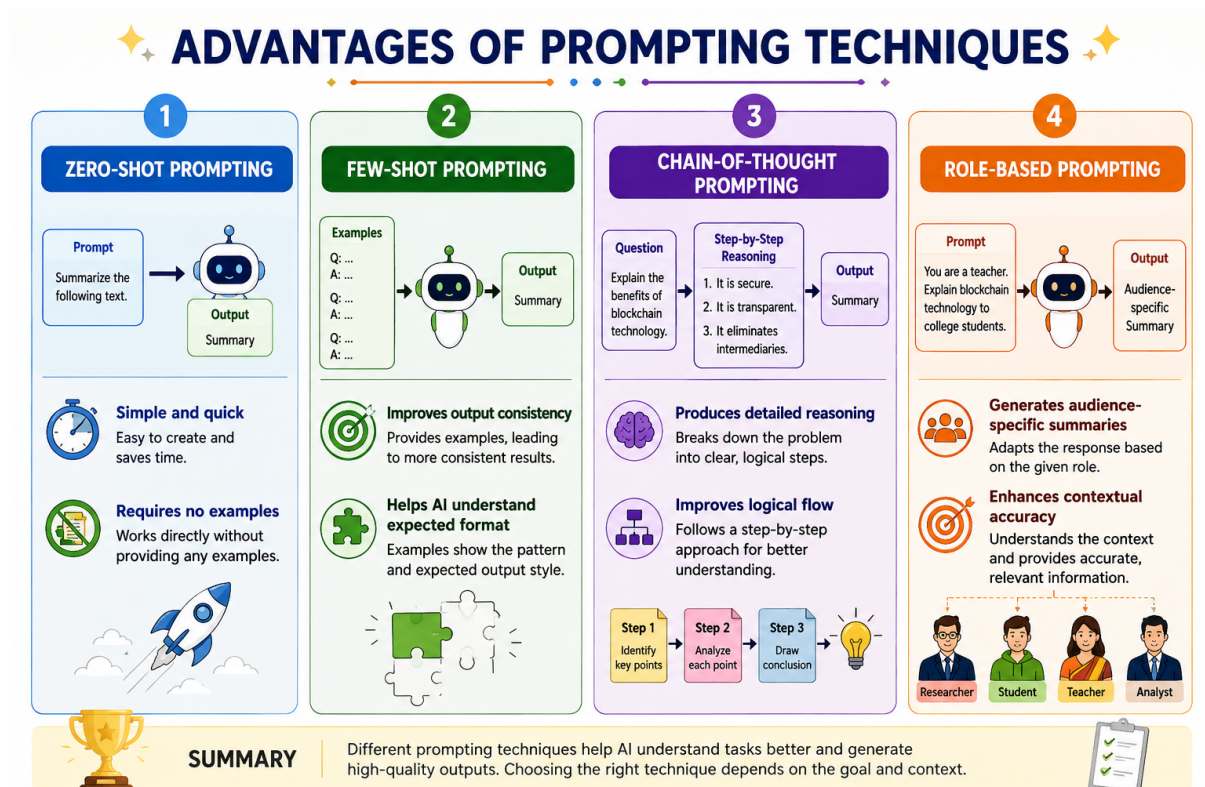
Parameter	ChatGPT	Gemini	Copilot
Accuracy	Highly accurate	Accurate	Moderately accurate
Coherence	Excellent flow	Good flow	Simple flow
Simplicity	Easy to understand	Moderate simplicity	Very simple
Speed	Fast	Very fast	Fast
User Experience	Interactive and detailed	Clean and concise	User-friendly

Overall Comparison of Prompting Techniques

Prompting Technique	Best Performing Platform	Observation
Zero-Shot Prompting	ChatGPT	Produced clear and balanced summaries
Few-Shot Prompting	Gemini	Followed examples effectively
Chain-of-Thought Prompting	ChatGPT	Generated detailed and logical summaries
Role-Based Prompting	ChatGPT	Adapted well to the assigned role

Observations

- ChatGPT generated more detailed and coherent summaries.
- Gemini produced concise and technically accurate responses.
- Copilot generated simpler summaries suitable for beginners.
- Chain-of-thought prompting produced the most informative outputs.
- Role-based prompting improved contextual understanding across all platforms.



Result

Thus, different prompting techniques were applied and evaluated successfully across ChatGPT, Gemini, and Copilot platforms. The outputs were compared based on accuracy, coherence, simplicity, speed, and user experience. Among the platforms, ChatGPT provided the most detailed and coherent summaries, while Gemini produced concise responses and Copilot generated simple beginner-friendly summaries.

Conclusion

This experiment demonstrated that prompting techniques significantly influence AI-generated summaries. Chain-of-thought and role-based prompting produced higher-quality outputs compared to zero-shot prompting. Different AI platforms showed varying strengths in summarization tasks, highlighting the importance of selecting appropriate prompting methods for specific applications.

References

1. [OpenAI ChatGPT](#)
2. [Google Gemini](#)
3. [Microsoft Copilot](#)
4. [Blockchain Technology Overview by IBM](#)